EMERGING TECH HUB

POSITION PAPER

ON

Quantum Computing

Quantum

Disclaimer:

This paper presents the Flex Community vision of Quantum Computing and will be enriched on a regular basis. It aims to help everyone in foreseeing potential applications in their own specific context.

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I. Executive summary

A. Why ?

The interest and enthusiasm for Quantum Computing and related technologies is growing because the issues and benefits (proven or hypothetical) are enormous and extremely disruptive. The maturity of the field has not yet reached the threshold that allows a scaling up of scientific advances in industrial and economic reality. Nevertheless, the major players in the computing industry (especially in North America and China) are fighting hard to get ahead and gain a leading position.

In Europe, the awareness around the subject has accelerated in the past two years. In fact, to deal with the complexity/cost of the investments, institutional and private actors have organized themselves into chairs/consortia to be able to release quantum calculators according to a 10-year roadmap.

With regard to the state of the art, the field is accomplishing significant progress both in terms of hardware and software. Indeed, technology giants like IBM and Google continue to increase the power of their respective prototypes (around 50-70 qubits this year). On the software side, many technological stacks (languages, frameworks and patterns) have emerged (particularly at IBM and Microsoft), thanks to open source. This will facilitate experimentation and simulation of quantum use cases.

B. Challenges

First, there is the technological challenge for solution providers. Will these companies be able to create fault-tolerant quantum computers with enough qubits to be a real differentiator? And in which time frame?

The second challenge is on the side of user companies, and it is social. We will need to find people capable of transforming a real problem (in the financial world in our case) into a quantum algorithm. While ensuring that it gives a real advantage over an optimized classic solution.

The third one is about security. The power of quantum computing would make obsolete a lot of security solutions (those based on RSA and ECC cryptography). It will be mandatory to identify the impact for such risk.

C. Stakes

With such a disruptive technology, the major challenge for banking & insurance is to find the right level of investment to make in terms of resources (human and financial). The maturity of this technology is a long way off (10 to 15 years) and the outcome is not certain. However, it is essential to be able by that time, to identify our sectors that will be impacted by this technology and to be prepared to use it as soon as it is feasible. The challenge is to be a player in this disruption.

Source – kpmglab_au

II. Why this focus for the bank & insurance?

A. Quantum Computing definitions

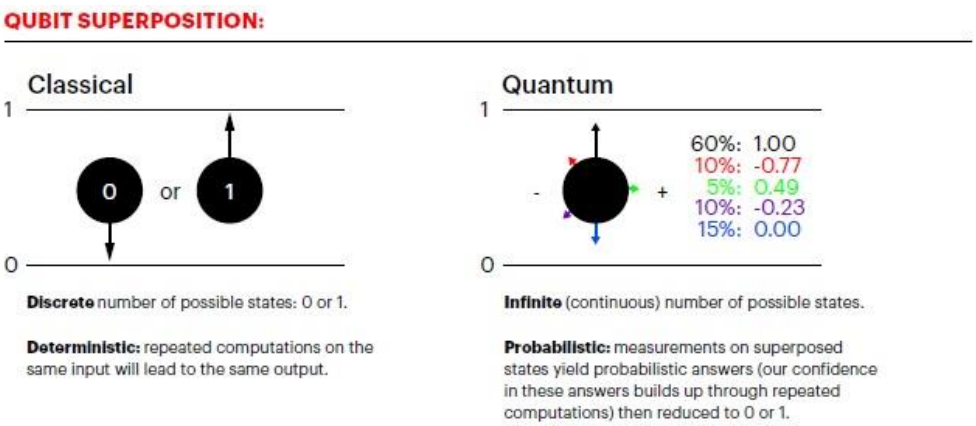
Conventional computers are based on transistors and require data to be encoded into binary digits (bits), which can only exist in one of two states at a given time (0 or 1). But Quantum computers use quantum bits (or qubits), which can exist in multiple states simultaneously. As a result, operations on qubits can amount to a large number of calculations in parallel, which could make certain kinds of computing problems much faster to solve.

Classical Computer	Quantum Computer
Basic unit : **bit** (1 or 0)	Basic unit : **qubit** (1 and 0 at the same time)
Deterministic – reliable, always the same answer	**Probabilistic** – gives a range of outcomes, not always the same answer
Based on transistors and semiconductors	Several technologies
Highly miniaturised, reliable	**Large**, cannot run for long
Easy to enter data	Radio frequency pulses
Easy to read data	Reading data can alter results
Runs at **ambient temperature**	Runs at very low temperature, **near absolute zero** (colder then universe)
Poor at finding prime numbers	Very good at **finding prime numbers** (Shor)
Good at sorting numbers	Much, much faster at **sorting numbers** (Grover)
Perfect for calculus	Perfect for **complex simulations**

Source – Morgan Stanley Reasearch

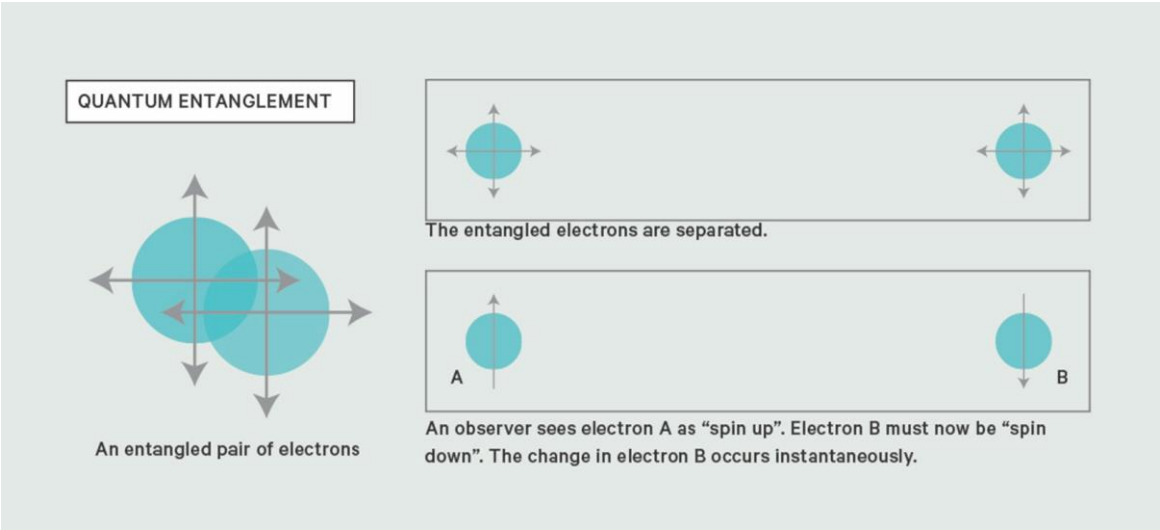
There are 4 major foundational concepts in Quantum Computing:

1. **Superposition**: In classical computing data are encoded into binary digits (bits), each of which is always in one of two definite states (0 or 1). However, Quantum Computing uses quantum bits (qubits), which can be in multiple states (0 and 1) simultaneously. This is known as quantum superposition.
2. **Qubit superposition:**



Source: Accenture - https://fstechadvisory.accenture.com/what-is-quantum-computing/fig-1-1-qubit-superposition/

3. **Entanglement** gives Quantum Computing the ability to scale exponentially. If one qubit simultaneously represents two states (superposition), two qubits represents four states when coupled together. They can no longer be treated independently; they now form a coupled, or entangled, super state. As more qubits link together, the number of states exponentially increase, which could lead to a computer with astronomically large computing power. As soon as you make a measurement on one of the entangled qubits, immediately, the system collapses, and the entanglement property disappear.



Source: https://sciencesprings.wordpress.com/2017/10/09/from-cosmos-einstein-bohr-and-the-origins-of-entanglement-wonderful-article-on-this-debate-between-einstein-and-bohr/

4. **Fragility**: Quantum states are quite fragile. If you measure, observe, touch, or perturb any of these states, they collapse to a classical state. The states don't stick around for very long, which is why it is difficult to build them for the moment.

5. **No cloning**: A corollary to fragility is the "No Cloning Theorem". The information entangled within a set of qubits will be lost if someone tries to observe or copy them. A quantum state cannot be copied without the sender or receiver realizing this.

In 1996, David DiVincenzo listed five key criteria for creating a Quantum computer:

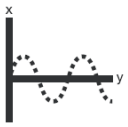
- A quantum computer must be scalable.
- It must be possible to initialize the qubits.
- Good qubits are needed, the quantum state cannot be lost.
- We need to have a universal set of quantum gates
- We need to be able to measure all of those qubits

And for quantum communication (transmitting intact qubits from place to place), two additional criteria must be satisfied:

- The ability to interconvert stationary and flying qubits, and
- The ability to transmit flying qubits between distant locations.

There are 2 kinds of Quantum Computers:

- Universal: based on Quantum logic circuits, built from quBits. It is possible to program it to solve almost every problem
- Specialized: built to solve only one kind of problems (like travel optimization).

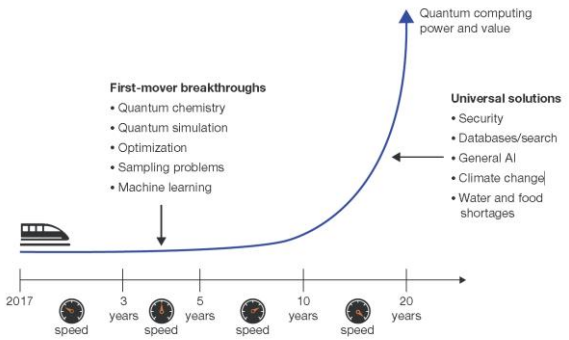
	What	How	Challenges	Who
Scalable, universal:	A general-purpose computer that can be programmed to solve many types of problems	Quantum logic circuits built from error-corrected, logical qubit gates	• Scaling hardware • Improving coherence time • Error correction • Software stack	IBM, Microsoft
Specialized: 	A specialized computer that can implement quantum algorithms for domain-specific problems	• Small-scale (noncorrected) quantum logic • Analog quantum simulation • Adiabatic annealing	• Finding the right problems • Designing algorithms • Improving qubit coherence and gate fidelity	D-Wave Systems, Google, IBM

Source – Forrester: A first look at Quantum Computing

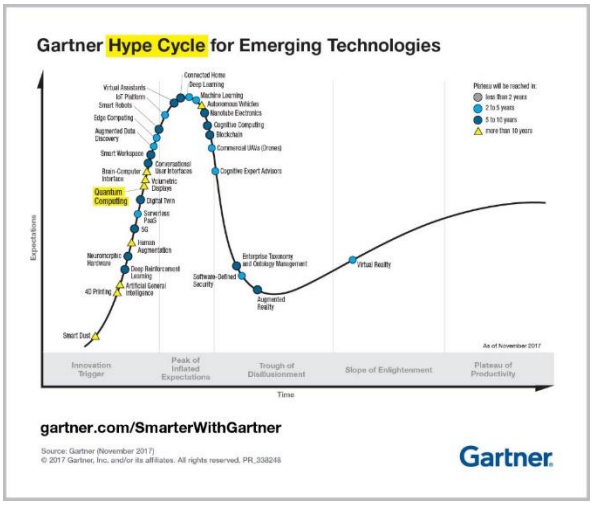
There are 2 kinds of Quantum Computers: Universal and Specialized.

B. Quantum Computing timeframe

According to Forrester, Gartner, IBM or university researchers, the time frame to have an error free universal quantum computer is **10 to 15 years from now**.

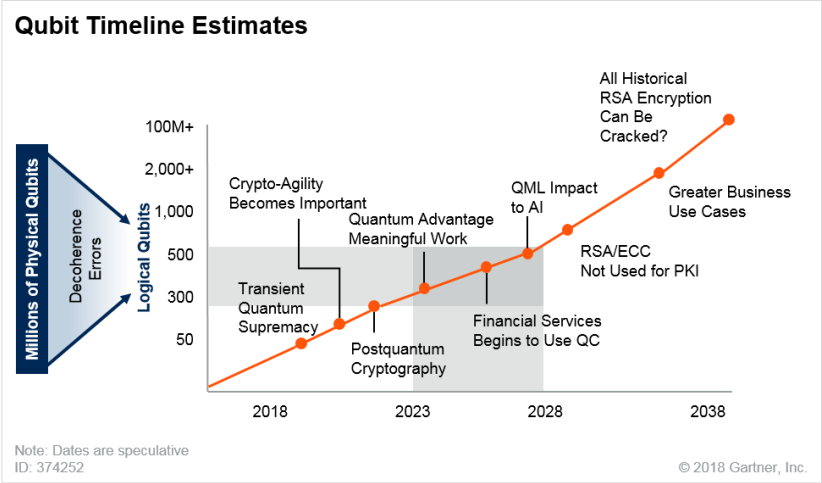


Source : Forrester : A fisrt look at Quantum Computing



Source :Gartner : Hype Cycle for Emerging Technologies

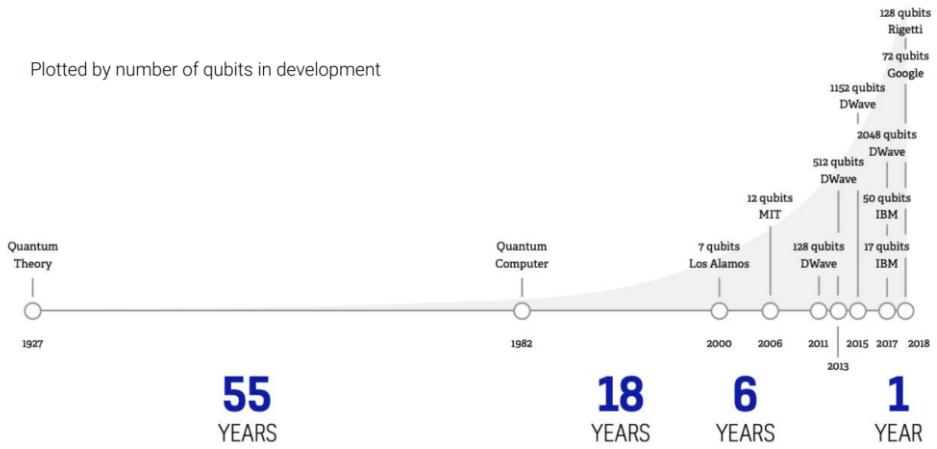
An estimated timeline has been given by Gartner, based on the number of logical quBits (a logical quBit is a group of hundreds to thousands of physical quBits, which are needed to be fault tolerant). As a reminder, the current usable quantum computers have “only” 50 physical non-error free qubits (50 for IBM and 53 for Google).



Source: Gartner

In the meantime (**in 3 to 5 years**), there could be a potential use for Noisy Intermediate-Scale Quantum (NISQ) computers (these are Quantum Computers which are not purely fault tolerant) which could bring an advantage compared to classical computing for specific use cases, which have yet to be identified and validated.

Looking backward, the field of quantum computing is evolving at the same rate as all other technologies (exponential acceleration).



Source: StranegWorks

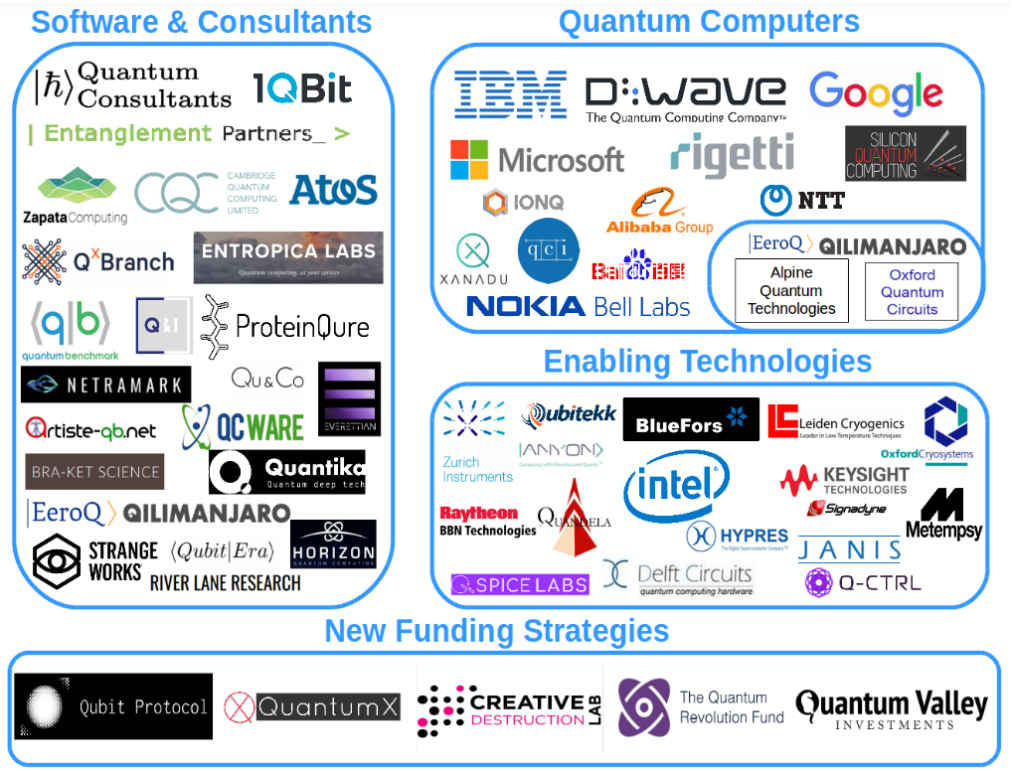
A real fault tolerant Quantum Computer is not expected before 10 to 15 years (2028 – 2033). In the meantime (3 to 5 years) Noisy Intermediate-Scale Quantum (NISQ) computers could be available for running real business cases.

C. The Quantum Computing ecosystem

The ecosystem is made by the major tech players (IBM, Google, and Microsoft), universities and startups.

The existing ecosystem of quantum can be considered as seperate two parts – hardware providers and software/solution providers. Some companies are active in both the hardware and the software domains (IBM, Microsoft, Rigetti, D-Wave, Google).

Even if we see no Chinese company in this picture, it is known that Alibaba and some Chinese universities are very active in Quantum Computing (In 2017 China announced that it was set to open a National Laboratory for Quantum Information Sciences by 2020. This included a 92-Acre, $10 Billion quantum research center).



Sources: Quantum World Association - 2018

Except for D-Wave which also offers their quantum computer for sale, the economic model of quantum computer manufacturers is to offer an access service through the cloud.

To do so, they offer cloud services to run algorithms on the physical quantum computers they host in their data centers / laboratories (Quantum Computer as a Service: QCaaS). As quantum computers require complex refrigeration devices, there is no benefit in having this type of computer in regular data centers.

At present, the 2 companies that offer the most mature environments (number of qubits, technology, languages for executing the first quantum algorithms for business use are **D-Wave** for its specialized quantum computer and **IBM** for its universal quantum computer.

Concerning the simulation of a quantum computer, we have **ATOS** (with the Quantum Learning Machine) for universal computing and **Fujitsu** (Digital Annealer) for specialized computing.

Microsoft and **AWS** have announced a cloud service to access simulators and physical quantum computers. For now, it is only available as a preview. The difference with IBM, is that it will be

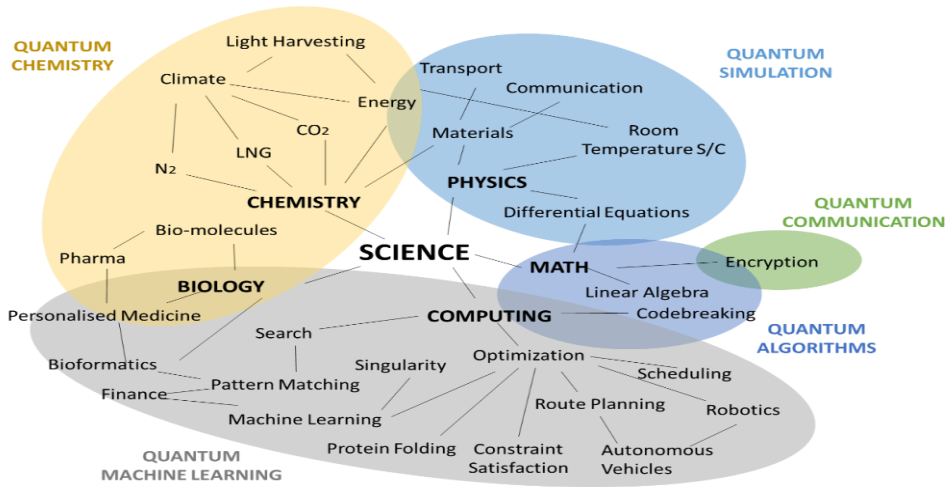
possible to have access to quantum computers made from different manufacturers and technologies.

D. Quantum Computing usage

Quantum Computing is best suited for solving problems using three types of algorithms: **optimization**, **sampling** and **machine learning**. These problems can be found in all technical domains.

The quantum domain, has been split into 5 verticals:

- Quantum Chemistry
- Quantum Simulation
- Quantum Communication
- Quantum Algorithms
- Quantum Machine Learning



Source – World Economic Forum, HSRC

For banks and insurances, the verticals which could bring some disruption are:

- Quantum Algorithms: with the usage of new linear algebra algorithm to improve *study financial trading strategies, forecasts and risk metrics*
- Quantum Communication: with the usage of quantum solutions to encrypt data and/or communication exchanges and offering the highest level of security
- Quantum Machine Learning: with optimization solutions applied to all possible areas, and the improved processing of ever-increasing amounts of data.

Quantum Computing Will Transform Almost Every Aspect of our Technology, Science, Economy & Life.

E. Quantum Computing and CSR

Quantum computing is a great opportunity to help reaching multiple sustainable goals defined by the United Nations.

Source: https://sustainabledevelopment.un.org/

According to a Zapata blog post (https://www.zapatacomputing.com/lets-make-quantum-computing-about-sustainability/), Quantum computing is bound to have a direct impact on at least 5 of them:

1. Zero Hunger (Goal 2) – new algorithms for nitrogen fixation
2. Good Health and Well-Being (Goal 3) – faster and cheaper drug discovery
3. Clean Water and Sanitation (Goal 6) – better water optimization and catalysts
4. Affordable and Clean Energy (Goal 7) – new batteries
5. Climate Action (Goal 13) – improve carbon capture, optimize fertilizer production

1) Quantum computing and climate change

- Power consumption and Greenhouse Gaz emissions:

Quantum Computing, by construction, should be more energy efficient then classical computing, as it has an exponential power scale, and in this way have a positive impact on climate change. But for now, it is very difficult to estimate this impact as we have no real business application on quantum computers. For the more common quantum computers, the most part of power consumption goes toward running the refrigeration unit (-273°C). The quantum processor itself requires a comparative pittance. "The operation of the quantum processor itself requires remarkably little power—only a tiny fraction of a microwatt—which is essentially negligible in comparison to the power needs of the refrigerator and servers," says Colin Williams, director of business development & strategic partnerships at D-Wave Systems. And adding another qubit, which in principal double the power of the Quantum Computer, do not need to double the refrigerator capacity. *"If a quantum device can solve a*

problem with much better [time to solution] scaling than classical computing, it would also win on power," says Matthias Troyer, a computational physicist at ETH Zurich.

Anyway, a study was published (Establishing the quantum supremacy frontier with a 281 Pflop/s simulation - https://iopscience.iop.org/article/10.1088/2058-9565/ab7eeb) after the Google quantum supremacy success (Quantum supremacy using a programmable superconducting processor https://www.nature.com/articles/s41586-019-1666-5) which try to demonstrate the energy efficiency of the quantum computer. With the figures from Google experience, they calculate that doing the same calculation on 2 different supercomputers (Electra, Summit) would consume much more energy then what has been consumed by Google's quantum computer Sycamore. And the difference is huge: 97 MWh for Electra, 21 MWh for Summit, and only 0.42 KWh, that's 50,000 to 230,000 less!

Another benefit is that Quantum Computers need less "Rare-Earth Elements" for their building and compare with the power of the Quantum Computer. And this is very important as the "Rare-Earth Elements" are vital to modern technologies and lifestyles (https://pubs.usgs.gov/fs/2014/3078/pdf/fs2014-3078.pdf).

- Other positive environmental impacts:

There are multiple domains in chemistry where Quantum simulation could bring an advantage and have positive environmental impacts.

A catalyst for "scrubbing" carbon dioxide directly from the atmosphere could be a powerful tool in tackling climate change. Although CO2 is captured naturally, by oceans and trees, CO2 production has exceeded these natural capture rates for many decades. The best way to tackle CO2 is not releasing more CO2; the next best thing is capturing it. "While we can't literally turn back time, [it] is a bit like rewinding the emissions clock," according to Torben Daeneke at RMIT University.

There are known catalysts for carbon capture but most contain expensive precious metals or are difficult or expensive to produce and/or deploy. "We currently don't know many cheap and readily available catalysts for CO2 reduction," says Ulf-Peter Apfel of Ruhr-University Bochum.

Given the infinite number of candidate molecules that are available, we are right to be optimistic that there is a catalyst (or indeed many) to be found that will do the job cheaply and easily. Finding such a catalyst, however, is a daunting task without the ability to simulate the properties of candidate molecules.

And that's where quantum computing could help. (https://www.weforum.org/agenda/2019/12/quantum-computing-applications-climate-change/)

We might even find a cheap catalyst that enables efficient carbon dioxide recycling and produces useful by-products like hydrogen (a fuel) or carbon monoxide (a common source material in the chemical industry).

Another example is regards the production of fertilizer. The actual process, so called Haber-Bosch process, consume approximatively 2-3% of global CO2 emissions. At the same time, many bacteria found in nature perform the same chemical reaction as the Haber-Bosch method does – allowing the fixation of nitrogen from the air to take place. Quantum computers have natural ability to simulate chemical reactions, like those done by the bacteria.

This will help to find new process, less energy-hungry which will produce fertilizer with a better environmental impact and helping to reduce hungry in the world.

Another domain concerns the battery. As the new world will be more and more "electrical dependent", we need also better batteries. There are a lot of car manufacturer (IBM with Mercedes/Daimler : https://www.ibm.com/blogs/research/2020/01/next-gen-lithium-sulfur-batteries/; Volkswagen with Google and D-Wave: https://media.vw.com/releases/1037) who are working on battery optimization with the help of Quantum Computing. *"We are working hard to develop the potential of quantum computers for Volkswagen. The simulation of electrochemical materials is an important project in this context,"* says Florian Neukart, Principle Scientist at Volkswagen's CODE Lab in San Francisco. In the long term, the quantum algorithm could simulate the chemical composition of a battery on the basis of different criteria such as weight reduction, maximum power density or cell assembly, and provide a design which could be used directly for production. This would help significantly accelerate the battery development process, which has been time-consuming and resource-intensive to date.

Car traffic and commute time are also domains where quantum computing will bring positive impacts. Reducing the traffic and commute time will reduce greenhouse gaz emissions, but also have positive impact on well-being of people. There are multiple experimentations done by different players, applying optimization quantum algorithms. And the results are very promising.

https://medium.com/@ford/mass-navigation-how-ford-is-exploring-the-quantum-world-with-microsoft-to-help-reduce-congestion-a9de6db32338

https://www.volkswagenag.com/en/news/stories/2018/11/intelligent-traffic-control-with-quantum-computers.html

2. Quantum computing and social impact

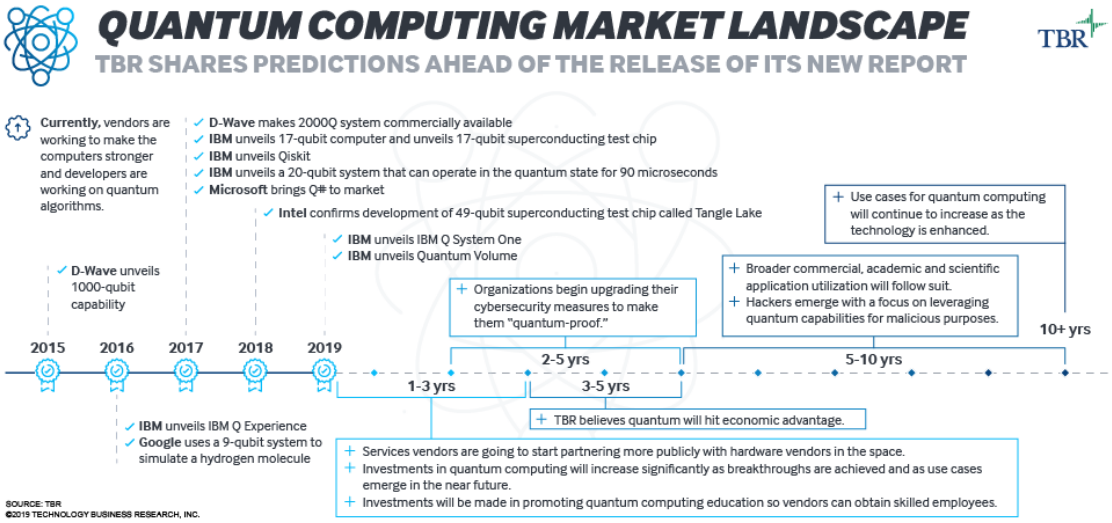
All climate change positive impacts have indirect positive social impacts (reduce hunger in the world, better life, reduce climate immigration, …). In additional to all this, Quantum Computing **could be a changer in Heath Care**. Over the last two decades, advancements in medicine and biomedical research have been vastly improved thanks to the continuous increases in computer processing. With the power of **Quantum simulation**, we could **create personalized medicine** for everyone on the planet taking into account lifestyle, genetics, and environmental factors.

But ethical aspects could be more negative. In destroying much of classical cryptography, quantum computers reduce online privacy and make it harder to hide things (both for good and for bad purposes). If access to quantum computers is limited to one or few governments, it could upset the balance of power between different countries; and if this access is limited to a few big companies it could lead to monopolies or oligopolies, increasing inequality in society.

III. How far Quantum Computing could reshape banking & insurance activities

Quantum Computing is coming with opportunities and threats:

- **Opportunities: the power will help banking and assurance in providing more accuracy in portfolio management, market simulation and fraud detection**
- **Threats: the cryptography used to secure the transactions and communication would be easily broken by quantum power**



According to Technology Business Research, quantum computers could bring economic advantages in 3 to 5 years and that companies should be prepared for the security issue (cybersecurity) in 2 to 5 years.

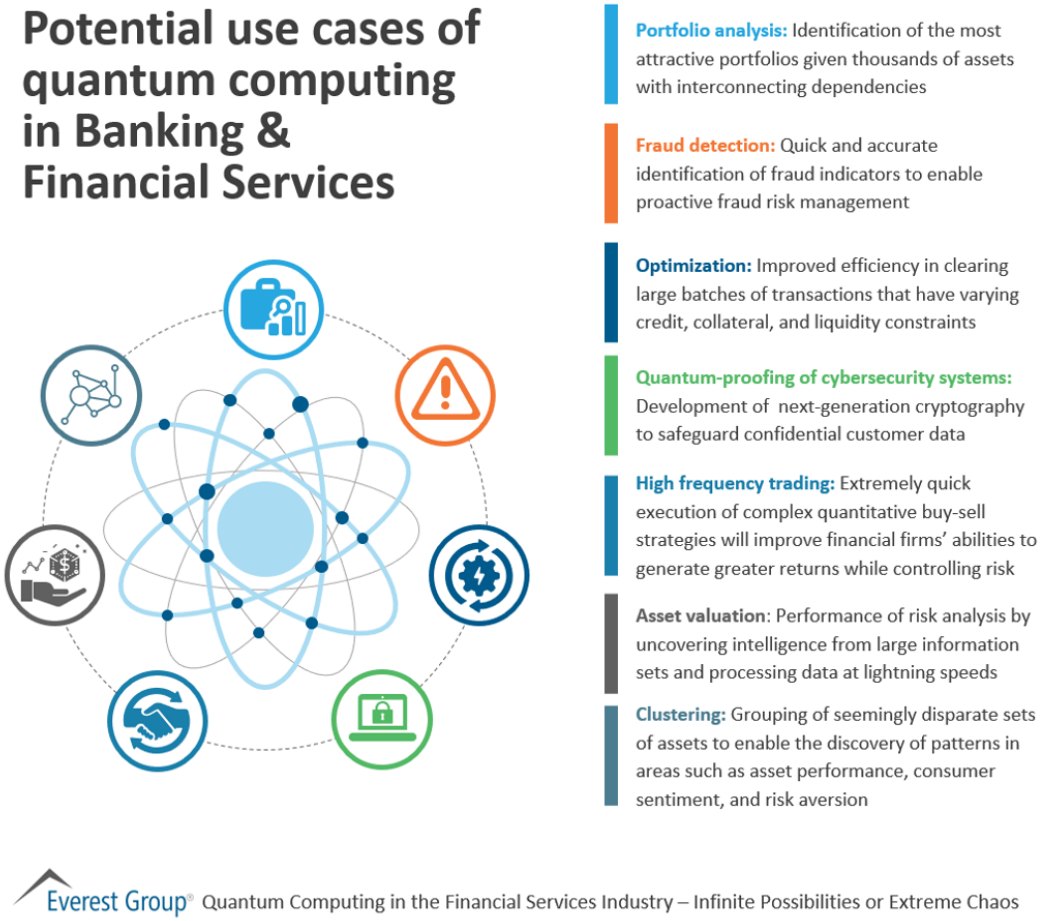
A. Opportunities:

As with other industries, the age of the quantum computer will have an impact on how to become a banker or insurer. The processing power of such a system will allow us to imagine new services for our customers.

The areas that will benefit from this power are those that have to deal with an increasingly important mass of data:

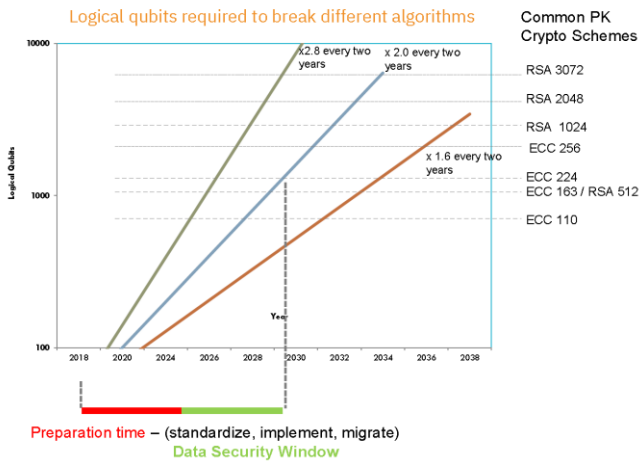
- Optimization: portfolio, credits & loans, ...
- Fraud: identifying signs of fraud ever earlier

In addition to these identified cases, other opportunities will be identified only after the arrival of this technology, when access to it will permit work to begin concretely on the subject (see the evolution of the Internet usage).



B. Threats:

The power of Quantum Computing will be an issue for security as it has been verified that some cryptographic algorithms would be broken (Shor's algorithm). The most vulnerable are those most used today (RSA, ECC).



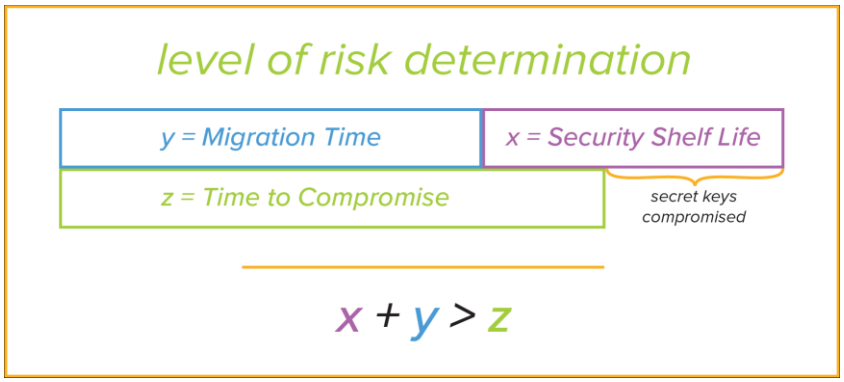
Source – IBM

Another amplifier, is that the quantum threat comes just as the internet of things and 5G mobile connectivity are arriving, with higher security requirements.

It will be necessary to make a risk assessment. It will help us to identify the potential systems to be adapted with a new security model. This should be mandatory to avoid security breaches, which could undermine the trust our customer have in us.

A particular attention has to be put on the time frame, because changing core security elements in a global infrastructure could need a long time. This is even truer if there is an impact on the way our customer interacts with us.

Professor Michele Mosca, co-founder and deputy director of the Institute for Quantum Computing at the University of Waterloo, put this threat in a mathematical formula to determine the level of risk.



Form him, it is essential for companies to begin transitioning now. Ideally, every organization should work to achieve quantum-proof coverage by 2025. “If x+y > z, you have a serious problem today !” (Mosca’s inegality).

IV. Challenges

For now, the challenge is having sufficient skills in identifying the real added value as well as the issues and risks of quantum computing for banking and insurance.

A. Prerequisites to experimentations

The world of quantum is very different from what we are accustomed to understanding. It is essential to think differently. There is a supplementary challenge in the implementation of experiments based on business applications. The skills needed for this are rare and mainly found in research centers specializing in quantum computing or in companies offering this type of computer (physical or simulators). We are speaking here about specialized mathematicians.

B. How to choose our partners

Considering the prerequisites mentioned, the choice for partners, should be focused on 3 types of partners:

- The world of university research: A partnership with a university should allow us to work on specific use cases and offer access to early information about the status of the research.
- A market leader in the creation of quantum computers: This should help to identify as early as possible the technological advances in the field of quantum computers, provide guidance in the development of differentiating uses cases, as well as helping build the trajectory to integrate this technology in the information system of the bank.
- A market leader in quantum computer simulation: In this case, it is about training / preparing collaborators to this universe. It should allow us to think differently about the problems and perhaps to find new methods applicable to classical computing.

V. Focus on use cases to determine Quantum Computing's impact

There are a wide range of business use cases identified which could impact banking and insurance but attention should also be given to post-quantum cryptography.

A. Business use cases

The power of quantum computer should bring added value in a number of use cases where optimization and/or machine learning is key:

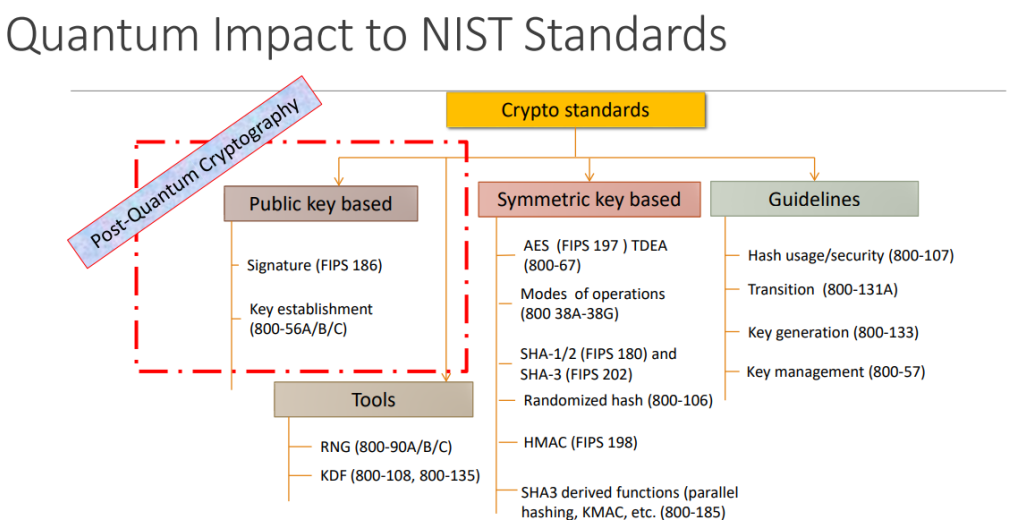
- Investment Portfolio Optimization
- Credit and loans scoring and classification
- Optimal trading
- Bank customer segmentation
- Reinsurance optimization
- Capital management
- Risk modeling
- Fraud detection

Some of these use cases have already been tested on quantum machines like D-Wave. There are some research documents available explaining the results :

- Financial Portfolio Management (Abu Dhabi securities Exchange)
- Optimal trading trajectory problem

B. Security use cases

Quantum computing could have a real impact on security, as it should be possible to crack actual cryptography keys based on RSA and ECC (Shor's Algorithm).



Source: The impacted cryptographic algorithm by quantum computing

Based on this issue, the NIST (National Institute of Standards and Technology) has asked for proposals on cryptography algorithms that would be quantum resistant (Post-Quantum Cryptography).

In December 2016, the NIST (National Institute of Standard and Technology) start a call to the public to submit post-quantum algorithms that potentially could **resist a quantum computer's** onslaught.

The agency spent one year collecting the submissions and another working with the larger cryptography community on a first round of review to focus on the most promising algorithms. Of the 69 submissions NIST received, these **26 algorithms made the cut.** (https://csrc.nist.gov/projects/post-quantum-cryptography/round-2-submissions)

- 17 Second-Round Candidate concerns public-key encryption and key-establishment algorithms
- 9 Second-Round Candidate concerns digital signatures

In July 2020, the result of the 3rd round (https://csrc.nist.gov/Projects/post-quantum-cryptography/round-3-submissions) has been published. The 15 remaining, have been split into 2 categories, the finalists and the alternate candidates.

- 4 Finalists and 5 Alternate Candidates for Public-key Encryption and Key-establishment Algorithms
- 3 Finalists and 3 Alternate Candidates for Digital Signature Algorithms

The goal is to have draft standards of the winning algorithms for **2022/2024**.

It would be valuable to have a review of the impact of the implementation of this new kind of security solutions.

VI. Conclusion

While waiting for the availability of fault-tolerant quantum computers (10 to 15 years or more) and to benefit from their unmatched computing power, preparation is needed Indeed using such computers is totally different from actual technology. For this, it is necessary to have people trained and to foster relations with universities and top partners.

It would also be useful to start working on business use cases that can be used on Noisy Intermediate Scale Quantum (NISQ) computers (these computers are only ~90% accurate), since the forecast for seeing such computers on the market is on a short / medium term horizon (3 to 5 years).

This can become a differentiating element for new services to the bank's customers. As an example, BBVA (Banco Bilbao Vizcaya Argentaria) has put in place a team dedicated to quantum computing since 2018 and already created 6 POCs with partners (Accenture, D-Wave, Zapata Computing, …).

Finally, regarding the issue of security, if quantum computers are able to break the encryption keys widely used in our systems to secure communications (SSL / TLS) and data encryption (public / private keys), it should be needed to invest time in understanding the impacts of post-quantum cryptography (PQC). This, in case it will be necessary to replace the current security solutions, which will necessarily require a lot of time (example: changing the method of encryption of all data requires the review of all applications using this data, but also the re-encoding of all information using new encryption algorithms.

APPENDIX

Richard Feynman: https://en.wikipedia.org/wiki/Richard_Feynman

DiVincenzo criteria's : https://en.wikipedia.org/wiki/David_DiVincenzo

https://www.microsoft.com/en-us/quantum/technology

https://newsroom.intel.com/news/intel-sees-promise-silicon-spin-qubits-quantum-computing/

https://iq.intel.com/readying-quantum-computing-lifes-biggest-mysteries/

https://ec.europa.eu/digital-single-market/en/policies/quantum-technologies

Quantum Algorithms (Ronald de Wolf)

Quantum Computing Market – Global forecast to 2023 (www.marketsandmarkets.com)

Quantum Computing – weird science or the next computing revolution? – Morgan Stanley

The Quantum Zoo: There are quite a few known quantum algorithms, and it can be difficult to get an overview of the field. Luckily, Stephen Jordan (at NIST) has assembled a list of brief descriptions for the quantum algorithms which are known so far. https://math.nist.gov/quantum/zoo/

Shor's Algorithm: https://arxiv.org/abs/quant-ph/9508027v2

Internal Quantum Community hosted by the Emerging Tech Hub:
https://sbc.safe.socgen/groups/emerging-tech-hub